

Squid inspired elastomer marine coating with efficient antifouling strategies: Hydrophilized defensive surface and lower modulus

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ABSTRACT

In antifouling applications for the marine industry, low surface energy coatings entail turbulent water flow to release marine biofouling, which presents a substantial challenge for antifouling in the static situation. The traditional solution is to add environmentally friendly antifouling agents, but it has the problem of exhaustion. Therefore, the low surface energy elastic antifouling coating without antifoulants has high research value. Herein, inspired by soft body and epidermal mucus of squid, the stable polyvinylpyrrolidone (PVP) hydrophilic segments were introduced to modify the polydimethylsiloxane-based polyurethane (PDMS-PU), realizing low surface energy elastomer coatings with hydrophilized defensive surface and reduced elastic modulus (<1.1 MPa). In an aqueous environment, the tailored surface exposed sufficient stable hydrophilic segments, exerting excellent antifouling performance, which improved the anti-adsorption effect on biological proteins, bacteria (antibacterial rate 95.24%) and algae (cover rate <3%). The coating exhibited excellent marine antifouling performance within 150 days and also gave a new impetus to developing an eco-friendly and sustainable solution for no-antifoulant marine antifouling applications.

1. Introduction

Applying an anti-fouling coating is one of the most economical and effective methods to prevent marine organisms or fouling [1–4]. Taking into account the increasing emphasis on marine ecological environment protection, in the application of antifouling coatings, humans have gradually reached a tendency to use environmentally friendly low surface energy coatings that do not release toxic materials into the ocean and do not produce man-made pollution such as microplastics [4–6]. The low surface energy antifouling coating based on PDMS can reduce the force and interaction between fouling and substrate molecules, which is conducive to the release of biofouling [7,8].

However, in the early stage during the colonization process of marine organisms growing on the substrate surface, the interface energy between the hydrophobic surface and water is relatively high, which is not conducive to preventing the adhesion of the biological protein-based condition film [9,10]. Marine microorganisms such as bacteria and

diatoms then settled and colonized on the low surface energy marine antifouling coatings [11,12]. When there is no turbulent water flow to release the fouling, more macro marine organisms will settle on its surface, which limits the application of low-surface marine antifouling coatings in static environments [3,13].

Squid is common marine creatures with soft body, whose size ranges from 1 cm to 10 m [14]. Unlike whales and sea turtles, marine fouling organisms (barnacles, mussels, bryozoans, etc.) rarely adhere to the low elastic modulus surface of the squid. For example, the epidermis of cuttlefish has elastic modulus for only 0.5–1.0 Mpa. The squid successfully survived in the ocean by virtue of the efficient and flexible swimming method. They rely on low-modulus bodies and static-water bones to move in the vast ocean from thousands of meters deep to sea level. Because squid has better pressure resistance and flexible movement performance, it is widely used in the design of underwater robots. The flexible fin unit and connective tissue that simulate the squid fin structure can simulate the mechanism of the animal's storage and release of

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elastic energy during the movement process, which has promising applications in the field of soft robots. The researchers measured the properties of a typical squid (squid *Loligo pealei*) mantle and its effect on its sports performance, and measured its average elastic modulus of 0.7–0.9 MPa. In addition, although squids are widely distributed in deep and shallow seas, move slowly in most instances, their surfaces are not attached to fouling organisms as sea turtles. This may be attributed to the low modulus and mucus (hydrophilic polysaccharides, protein and bound water) of the squid's epidermis for its defence against fouling organisms [4,15,16]. Moreover, similar to most fish, the squid skin can secrete hydrophilic mucus which can insulate the epidermis from contact with fouling organisms such as bacteria and algae [16–18]. Barnacle larvae and mussels tend to attach to hydrophobic solids, thus, this hydrophilic mucus can "confuse" these organisms and prevent them from attaching to the epidermis [4]. However, there are relatively few reports on the application of the bionic soft skin of squid in the antifouling field. So inspired by squid, as shown in Scheme 1, hydrophilized defensive surface and reduction of surface modulus are feasible methods to enhance the marine antifouling performance of underwater equipment surface.

The hydrophilized low SE coating combines the respective advantages of hydrophilic and hydrophobic surfaces. As the main body of coating is made of hydrophobic polymer, the coating will not be swollen by seawater like hydrogels to be damaged easily by water flow. The hydrophilic segments can migrate to the coating-seawater interface and form a hydration layer on the surface, which is not conducive to the attachment or sedimentation of marine organisms [9,19–21]. One of the traditional hydrophilic-modifying materials is polyethylene glycol (PEG), which has been widely used in various hydrophilic modified materials [22–25]. Nonetheless, in harsh environments such as the ocean, PEG lacks long-term stability and is easily oxidized and degraded by microorganisms [26–29]. Therefore, subsequent research has focused on alternatives to hydrophilic segment PEG, such as zwitterions and polyelectrolytes, which can form a highly structured water layer in water [30–34].

Polyvinylpyrrolidone (PVP) has a flexible chain-like structure. The main polymer chain is rich in methylene and has lipophilicity. The lactam on the pyrrolidone ring has a strong polarity, which gives the material hydrophilicity. Among them, the oxygen atom tends to donate electrons to the pyrrolidone ring, so that the oxygen is positively charged, and the whole molecule is weakly cationic. The higher electron density on the ring gives it a stronger ability to form hydrogen bonds and form complexes in terms of surface activity [35]. PVP also has good

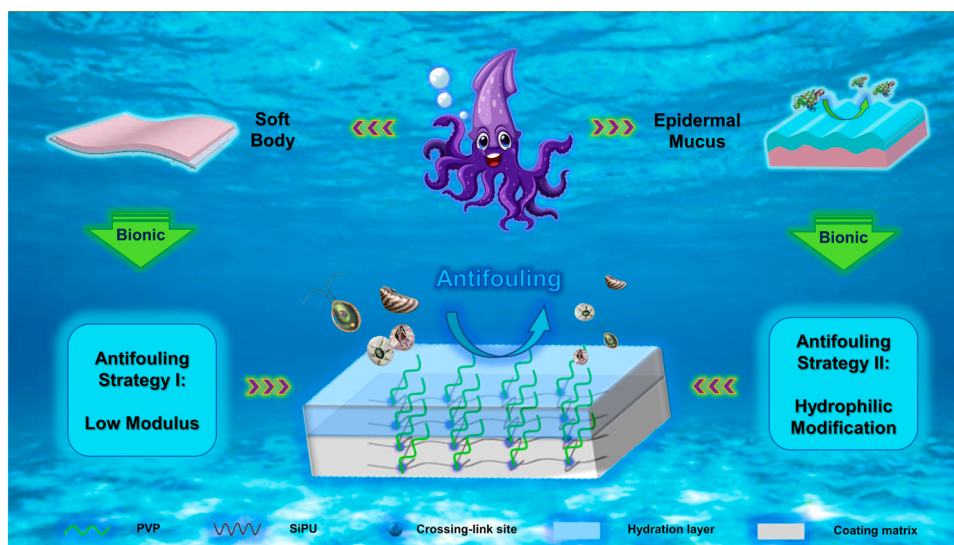
stability, for example, its structure will not be oxidized under the condition of 100 h in 60 °C, 3% H₂O₂ solution [36]. PVP can be closely combined with water molecules to form a stable hydration layer on the surface of the coating and eliminate the electrostatic force between the surface of the coating and molecules such as biological proteins [37]. Therefore, as reported previously, the coating has the potential to effectively resist the settlement of biofouling [10,15,38,39]. Besides, on account of hydrophilicity difference and phase separation, the PVP segments in the PDMS matrix could induce cavity-type vesicles in the coating body [39]. This inhomogeneity could bring in the reduction of elastic modulus of the coating, which has the potential to facilitate fouling release [23].

In this research, we reported a hydrophilic modification of the polydimethylsiloxane-based polyurethane (PDMS-PU) via the PVP hydrophilic segments. Benefiting from hydrophilicity and phase separation by PVP, the low surface energy coatings exhibited improved hydrophilicity and achieved reduced elastic modulus (<1.1 MPa) to enhance antifouling performance. Such a low elastic modulus was superior (decreased more than one magnitude order) to that of most of the previously reported PDMS-based polyurethane coatings [33,40–42]. The prepared antifouling polyurethane coating with tailored hydrophilic modification was tested in the sea area of Zhoushan, and its antifouling performance was carried out for 150 days. The work gave a new impetus to developing an eco-friendly and sustainable solution suitable for no-antifoulant marine coatings.

2. Materials and methods

2.1. Materials

α,ω -hydroxypropyl-terminated polydimethylsiloxane (HT-PDMS, Mw=3000 g/mol) was purchased from Jiande Polymer New Materials Co., Ltd., and dried under reduced pressure at ~0.01 MPa, 60 °C for 3 h before use. Polytetrahydrofuran (PTMG, Mw=2000 g/mol, ≥99.9%) was purchased from BASF and used as received. Isophorone diisocyanate (IPDI, ≥99.5%) was purchased from Aladdin and used as received. Dibutyltin dilaurate (DBTDL, 95%) was purchased from Aldrich and used directly. N-vinylpyrrolidone (NVP, 99%) was purchased from Macklin and used after purification by distillation. 2,2'-azobis(2-methyl-N-(2-hydroxyethyl)propionamide) (AMPAHE, 98%) was purchased from Macklin and used as received. β -mercaptoethanol (MCE, 99% biotechnology grade) was purchased from Macklin and used as received. γ -Isocyanatopropyltriethoxysilane (IPTS, 95%) was purchased from



Scheme 1. Squid inspired antifouling methods of PVP-x-SiPU coating: lower modulus and hydrophilized defensive surface.

Macklin and used as received. Isopropanol (IPA, AR, $\geq 99.5\%$) was purchased from Macklin and used as received. Xylene (99%) was purchased from Macklin and used as received. Tetrahydrofuran (THF, analytical grade, AR) was purchased from Sinopharm Chemical Reagent Co., Ltd. and used as received. Phosphate buffer solution (PBS, pH 7.4) was purchased from Macklin and used as received. Artificial seawater is prepared according to ASTM D1141–98 (2003). The diatom species were purchased from Jiangxi Enyang Biotechnology Co., Ltd., including *Bacillariophyceae* (diatom *Navicula*, ~ 80 wt%, and *Coscinodiscus*, ~ 15 wt%) as well as other phyla (*Rhodophyta*, *Phaeophyceae*, *Chlorophyta*, *Chrysophyta*, and *Euglenophyceae*, < 5 wt%). Fluorescein isothiocyanate-labeled bovine serum albumin (BSA-FITC, 5 mg/mL) was purchased from Meilun Biotech, and diluted to 0.05 mg/mL with phosphate buffer solution when using. Deionized water is used to determine the wettability of the coating. Other reagents were used without further purification.

2.2. Preparation of monofunctional hydrophilic segment Hydroxyl-terminated Polyvinylpyrrolidone (PVP-OH)

The monofunctional hydrophilic segment PVP-OH was prepared by free radical polymerization. According to the method described in the literature, isopropanol and mercaptoethanol are used as chain transfer agents, and under the action of the macromolecular free radical initiator

AMPAHE, hydroxyl-terminated low molecular weight polyvinylpyrrolidone is synthesized by free radical polymerization. The molecular weight and polydispersity index of PVP-OH can be controlled by the ratio of the amount of reactants added. As shown in Fig. 1 A, the specific steps are:

(1) 30 g NVP (270 mmol), 0.7783 g AMPAHE (2.7 mmol), and 0.844 g MCE (10.8 mmol) were dissolved in 300 g IPA.

(2) After degassing with argon for 15 min, polymerize at 85°C for 24 h.

(3) After the reaction is completed, most of the IPA is removed by rotary evaporation at 80°C to obtain a lightly yellow colloidal polymer. The product was precipitated in 200 mL of ether, and the solid was collected, then dissolved in 15 mL of dichloromethane, and again precipitated in 200 mL of ether. Finally, the product was transferred to an evaporating dish and dried in a vacuum oven at 80°C overnight.

(4) The obtained PVP-OH is dissolved in tetrahydrofuran and prepared as a 50 wt% solution for later use.

More details like FT-IR spectrum and ^1H NMR of the prepared PVP-OH can be found in Fig. S1.

2.3. Synthesis of PDMS-based Polyurethane with tailored hydrophilic modification via PVP (PVP-x-SiPU)

To realize adequate adhesion with the substrate while maintaining

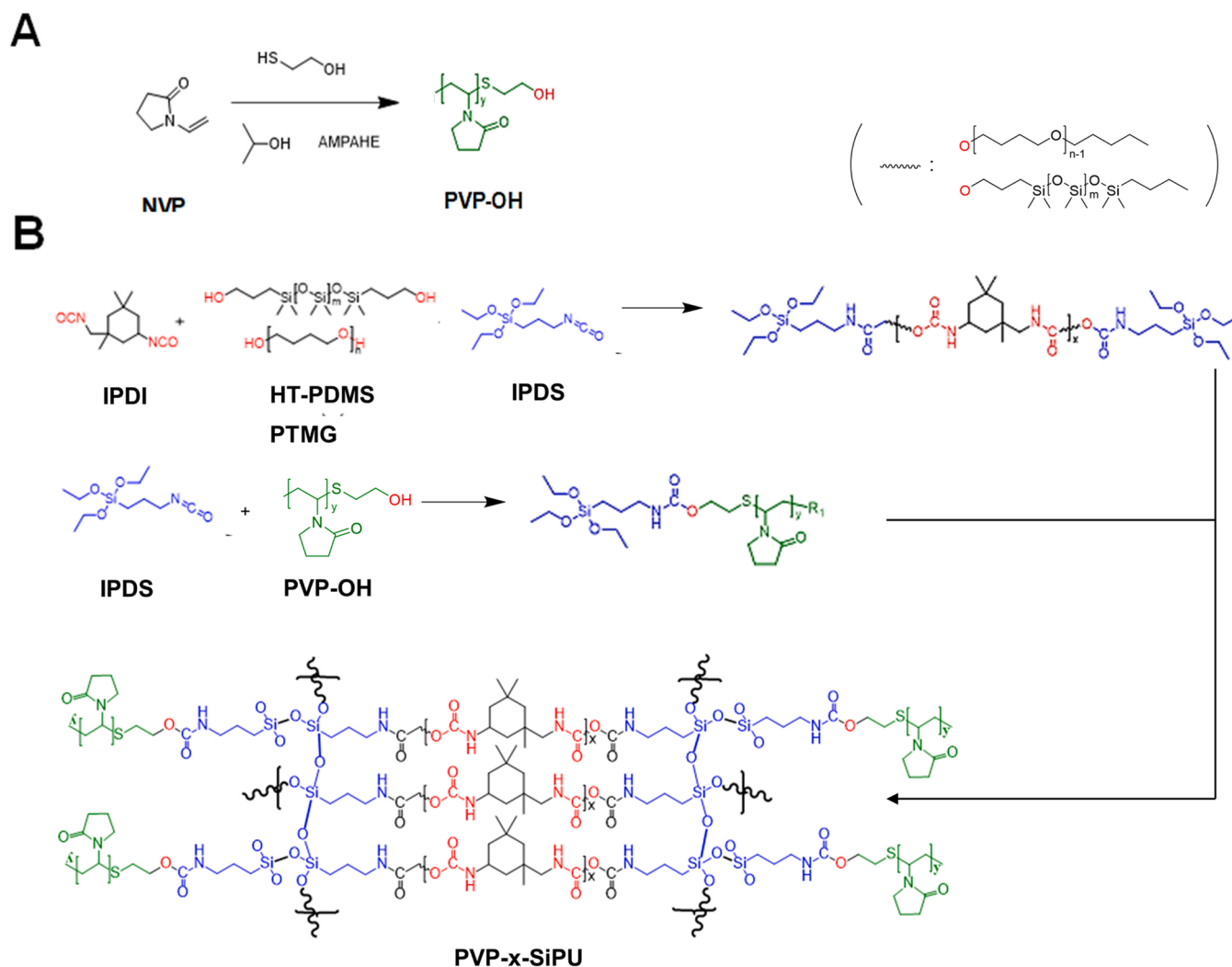


Fig. 1. (A) Synthesis of hydroxyl-terminated poly(N-vinylpyrrolidone) (PVP-OH) (B) synthesis of PDMS-based polyurethane with tailored hydrophilic modification via PVP (PVP-x-SiPU).

low surface energy characteristics, multiple soft segments (PDMS and PTMG in this research) with different polarities were taken as soft segments [43]. The PVP-OH has reactivity with the isocyanate group and is then introduced into the PDMS-based polyurethane through the covalent bond to form a stable affinity (Fig. S2. (B)).

HT-PDMS and PTMG and a small amount of IPDI were formed into a polyurethane prepolymer with excess hydroxyl groups via polycondensation reaction with the action of organotin catalyst, and then the hydroxyl-terminated polyvinylpyrrolidone PVP-OH solution was added and mixed well. Finally, add IPTS to cap the polymer (Fig. 1B). Different addition amounts of PVP-OH act as a gradient to form different coatings.

First, add a mixture of 3.75 g PTMG₂₀₀₀ ($M_w=2000$ g/mol, 1.875 mmol) and 5.625 g HT-PDMS ($M_w=3000$ g/mol, 1.875 mmol) into a three-necked flask with nitrogen inlet and outlet, and dissolve with a mixed solvent composed of 17 g xylene and 10 g tetrahydrofuran, under the protection of nitrogen, stir at 80 °C to form a homogeneous solution. Then 0.64 g IPDI and 0.05 g DBTDL were used as the hard segment and the catalyst and were added dropwise to the flask (OH: NCO=1.30:1), and the reaction was stirred for 2 h under a nitrogen atmosphere at 80 °C to obtain a transparent prepolymer solution. Subsequently, 50 wt% PVP-OH solutions with different mass gradients (0 g, 2 g, 4 g, 6 g) were added for hydrophilic modification. After stirring, the corresponding different mass gradients (0.5 g, 0.7 g, 0.9 g, 1.1 g) IPTS was used as a blocking agent and reacted at 80 °C for 2 h. For convenience, the hydrophilic modified PDMS-based polyurethane is expressed as PVP-x-SiPU, where x is the weight of PVP-OH added to the reaction system. The synthetic data of different coatings are summarized in Table S2. Due to the existence of the triethoxysilane group, the coating can be applied and moisture cure in situ at room temperature.

2.4. Coating Preparation

The slides were ultrasonically cleaned with acetone and ethanol for 10 min, then rinsed thoroughly with deionized water and dried. The prepared hydrophilic modified PDMS-based polyurethane solution was dropped with a pipette on the treated glass slide to form a coating with the same film thickness (with a dry film thickness of ~ 100 μm). It can also be dripped on the molds pre-coated with epoxy general anticorrosive primer for adhesion test or poured into clean PTFE molds to obtain a peelable coating film for storage modulus test. The PVP-x-SiPU coating was placed in a dry, ventilated room, and cured at room temperature for 24 h. The coating used for the marine anti-fouling test was applied on special sea test hanging panels with the size of 35 cm $\times$ 25 cm, pre-coated with epoxy general anti-corrosion primer (dry film thickness ~ 150 μm). The coating sample was coated on the hanging panels by air spraying or brushing with a dry film thickness of ~ 100 μm . The coating sample was slightly transparent, and the intuitive color of the hanging board after spraying is the primer color. The coating on the hanging panels should be cured for at least 2 days at room temperature before being immersed in the seawater.

2.5. Characterization

Fourier transform infrared (FTIR) spectroscopy data of the PVP-x-SiPU coatings in the range of 500–4000 cm^{-1} were performed by Nicolet 5700 FT-IR spectrometer. ^1H NMR spectra of the samples were recorded using a Bruker 400 MHz NMR spectrometer (Bruker Advance III 400 NMR) and carried out with 100 mg reaction solution (with the PVP-x-SiPU content of approximately 30%) in CDCl_3 at room temperature, due to the crosslinking of the polymer after drying. The molecular weight of PVP-OH was synthesized by Waters 1525/2414 gel permeation chromatography with tetrahydrofuran as the mobile phase, the flow rate of 1.0 mL/min, at 35 °C. The surface element distribution of the coating before and after response was characterized by SEM and its accompanying energy dispersive X-Ray spectrometer (EDS). The SEM used was field emission scanning electron microscopy (SU-8010, Japan),

and the mapping mode of EDS was used to characterize C, Si, O, N, and S elements. Each sample was purged with nitrogen before testing, and then gold was sprayed on the surface of the sample. The elemental composition of the coating surface was analyzed via X-ray photoelectron spectroscopy (XPS, Thermo Scientific, USA). The surface structure of the coatings was observed under field emission scanning microscopy (FESEM, Hitachi SU-8010) at an accelerating voltage of 3 kV. The contact angles (CAs) of deionized water and hexadecane on the surface of various coatings were measured with droplets (5 μL) through the sessile drop method through the CAM 200 optical contact angle goniometer (KSV Instruments Co., Ltd., Helsinki, Finland) at 25 °C. The average value of the statistics was obtained by measuring at least 5 different positions on the same sample, and the surface energy of the samples was calculated using the Owens-Wendt-Rabel-Kaelble method (Table S1) [44].

2.6. Lap shear tensile and adhesion tests

The organisms are supposed to be released by the shear forces turbulent water flow generated by ship movement, so the mechanical durability of prepared PVP-x-SiPU coatings was conducted by lap shear tests. As for lap shear tests, the process was performed to evaluate the adhesion strength of the coating to the substrate using a Zwick/Roell Z020 universal tester according to the method described in the article [45,46]. A sample was clamped at one end and pulled from another clamped end at a constant rate (6 $\text{mm}\cdot\text{min}^{-1}$), the tests were performed at room temperature, and five different points on each sample were tested to obtain an average value. The pull-off adhesion of the coatings to the epoxy-coated cylinders was using a Zwick/Roell Z020 universal tester according to the method described in the article [40]. The measurement area was 20 mm in diameter with the pull at 0.1 MPa/s. Five times of each sample were tested to obtain an average value.

Storage Modulus Test. The tensile tests were also carried out using a Zwick/Roell Z020 universal tester at a speed of 25 mm/min. At least five coating film samples were tested for each sample to obtain an average value of storage modulus.

2.7. FITC-BSA anti-adsorption, bacteriostatic, algae plate inoculation experiments and marine field antifouling test

FITC-BSA anti-adsorption test. The 18 mm $\times$ 18 mm coating samples were put into a six-well plate, 6 mL 0.05 mg/mL FITC-BSA solution was added to completely immerse the coating, and after incubating for 3 h in a dark environment, carefully remove the samples with a tweezer. Observe the coatings and take pictures under an inverted fluorescence microscope (XD, Sunny Optical Technology).

Bacterial Biofouling Resistance Test. The antibacterial performances of the coatings were investigated against *E. coli* with the plate count method, as described elsewhere [47]. Basically, aliquots of the 10^8 CFU/mL log-phase *E. coli* suspension were prepared in Luria-Bertani media. The coated coverslips (18 $\times$ 18 mm^2) were sterilized with high temperature and high pressure (121 °C, 100 kPa) for 15 min. The coating samples on coverslips were immersed in 5 mL of *E. coli* suspension (10^8 CFU/mL) and incubated in a shaking incubator at 37 °C for 5 h. After inoculation, the coating samples were then gently placed in tubes with 10 mL of Ringer's solution with tweezers, sonicated for 5 min, and vortexed for 30 s to collect the adherent bacterial cells on the coverslips. Six 10-fold dilutions were prepared, and the surviving bacteria were counted by the spread plate method (100 μL samples of the decimal dilutions were spread on a Petri dish that contained Luria-Bertani agar). The Petri dishes were incubated at 37 °C for 24 h. After incubation, the colonies were counted. The percentage reductions in bacterial colonies, as compared to an uncoated coverslip (as a blank control), were reported as the mean value of three replicate samples. Specifically, the bactericidal rate (E_b) was calculated by the following equation:

$$E_b = \frac{N_b - N_c}{N_b} \times 100\%$$

where N_b and N_c are the numbers of colonies (which means the surviving bacteria cells) corresponding to the uncoated coverslip and functional coating samples, respectively.

Algae Plate Inoculation Test. Cultivate the marine diatom using methods described in the literature [6,48]. The media recipe was a standard f/2 culture medium with Na_2SiO_3 (7.3 mg L^{-1}) added as a source of Si. The culture was maintained in standard illumination conditions with a 12:12-h light/dark cycle (Philips fluorescent white) at 25°C . The diatoms used in the experiments were in the exponential phase of growth with density of $10^5/\text{mL}$. The coated glass slides were put into algae suspension at 25°C with the illumination of natural light. After 7 days, the biofilm was gently rinsed with seawater to remove unsettled algae and then observed under a microscope and quantified by coverage area percentage of the diatoms in microscopy images.

Marine Field Antifouling Test. To get the antifouling properties of hydrophilic modified PDMS-based polyurethane coating (PVP-x-SiPU), the coating samples on sea trial hanging panels were immersed in the Zhoushan sea area of the East China Sea at depths around 0.5–1.5 m for 150 days. The surface fouling state of the hanging panels was checked and recorded by the camera. The coverage rate of diatoms was based on statistical data. 10 locations were randomly selected on the surface of the coating soaked in the diatom culture medium, and the optical microscope was used to observe and take photos. The average coverage of the diatoms was calculated and the error bars were marked.

3. Results and discussion

3.1. Synthesis of PDMS-Based Polyurethane with tailored hydrophilic modification via PVP (PVP-x-SiPU)

In this research, three kinds of PVP-x-SiPU coating modified with PVP were prepared. As shown in Fig. S2. (B), The strong and broad absorption peak at $3255\text{--}3570 \text{ cm}^{-1}$ is recognized as the absorption peak of water in PVP. As the content of PVP in the coating increases, the broad peak absorption becomes stronger, and the absorption peak at 1700 cm^{-1} (C=O) is enhanced.

Due to the IPTS end-capping, the presence of ethoxy groups in the hydrophilic modified PDMS-based polyurethane antifouling coating can be directly cross-linked in a moisture-curable manner at room temperature. As shown in Fig. S4 and Table S3, the elemental analysis of EDS and XPS was used to characterize the synthesis of PVP-x-SiPU. Since N and S are characteristic elements of PVP-OH, the increase of N and S in the surface elements can indicate that PVP is successfully introduced into the polymer chain. The SiPU without PVP had few N element (from urethane bonds) and no S element, while PVP-20-SiPU had more N content (from PVP-OH and urethane bonds) and S content (from PVP-OH). The EDS mapping results showed that the height of the N characteristic peak and C characteristic peak of PVP-SiPU were significantly higher than those of SiPU. (Because the S content was relatively low, it was not marked by the software system of EDS.) This further showed that SiPU was successfully modified by PVP.

3.2. Characteristics analysis of the PDMS-based polyurethane coatings

As shown in Fig. 2, the surface morphology of the coating was analyzed by FESEM. Compared with the non-hydrophilic modified coating PVP-0-SiPU, there were $1\text{--}5 \mu\text{m}$ uniformly distributed bumps and cavities on the surfaces of PVP-10-SiPU, PVP-20-SiPU, and PVP-30-SiPU. As the amount of PVP-OH increased, the density of protrusions and holes on the coating surface increased. The results revealed that, during the cross-linking process, the hydrophilic segments PVP-OH were assembled into the PDMS-based polyurethane matrix to form micro-vesicles [39]. Because the PVP segment in the polymer network has hydrophilic parts, it will spontaneously aggregate under the action of hydrophilic & hydrophobic interactions and hydrogen bonds to form hydrophilic micro areas. With the increase in the amount of PVP-OH, the hydrophilic segments on the surface of the coating can be arranged more tightly, and the area of the hydrophilic and hydrophobic domains is denser, which is beneficial to construction of hydrophilic micro-area, and is not conducive for marine life to explore the attachment site of settlement to enhance the anti-fouling performance [9].

3.3. Surface wettability and surface energy test

The wettability of the coating can be reflected by the water contact

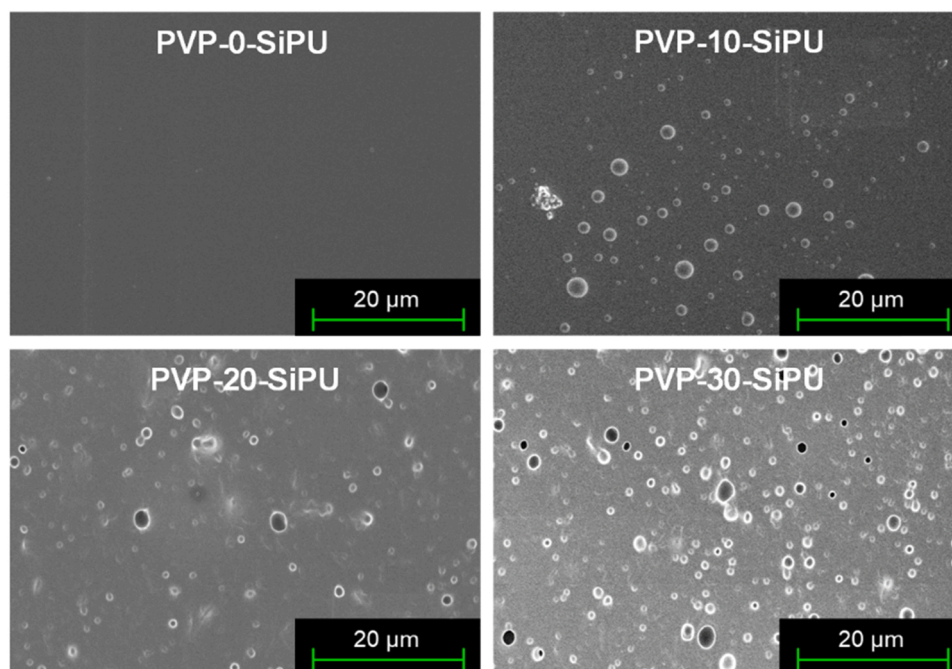


Fig. 2. The FESEM images of the coatings PVP-x-SiPU.

angles (WCAs). According to the contact angle of deionized water and N-hexadecane, the polar component and dispersion component of the coating were calculated by the Owens-Wendt-Rabel-Kaelble method, and then the surface energy of the coating was obtained (Table S1). The introduction of PVP can improve the hydrophilicity of the coating, and the degree of hydrophilicity is positively correlated with the amount of PVP added. The higher the PVP content, the lower the water contact angle of the coating. Due to the stronger polarity of PVP, the contact angle of hexadecane increased with the increasing amount of PVP. After being immersed in artificial seawater for three days, the WCAs of the PVP-0-SiPU coating without PVP decreased slightly, while the WCAs of the PVP-containing coating decreased, indicating that after immersion in artificial seawater, the hydrophilicity is further improved. The hexadecane contact angle of the coating that was immersed in artificial seawater for three days decreased with the increase of the PVP addition, which may be related to the formed hydration layer like mucus of cuttlefish epidermis. The existence of the hydration layer is conducive to the spread of non-polar liquid on the surface. These test results showed that the hydrophilicity of the coating surface had been improved, and the coating had the ability to form a hydration layer. According to the results of surface energy, after being immersed in artificial seawater, PVP increased the polar component of the coating. The overall surface energy of the coating increased with the increasing amount of PVP, however, the range is maintained between 22.5 and 34.0 mN/m. According to the literature, this surface energy range is still beneficial to inhibit the attachment of barnacles and other marine organisms [33,49].

3.4. Lap shear and coating adhesion test

The lap shear test and pull-off test can reflect the adhesion strength when the external force acts on the coating tangentially and normally.

Compared with the unmodified coating, the introduction of PVP-OH affects the homogeneous system of the original polyurethane system, resulting in a micron-sized vesicle structure, so the introduction of PVP-OH weakened the mechanical strength and adhesion of the coating and epoxy substrate. As shown in Fig. S3A, With the increase of PVP-OH, the adhesion of the coating slowly decreased, but the minimum adhesion can still reach about 0.3 MPa. The pull-off test reflects the direct adhesion strength between the coating and the substrate. With the further increase of the PVP-OH content, the adhesive force also decreases, while the adhesion strength was higher than 0.4 MPa (Fig. S3B). Although the induce of PVP-OH would weaken the mechanical properties of the coating, the adhesion strength was strong enough to withstand the

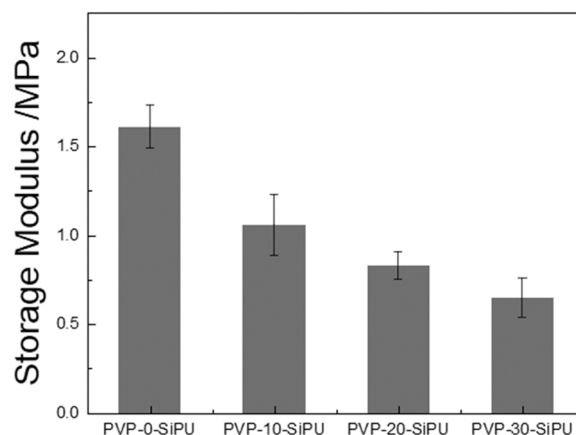


Fig. 4. The storage modulus of the coatings PVP-x-SiPU.

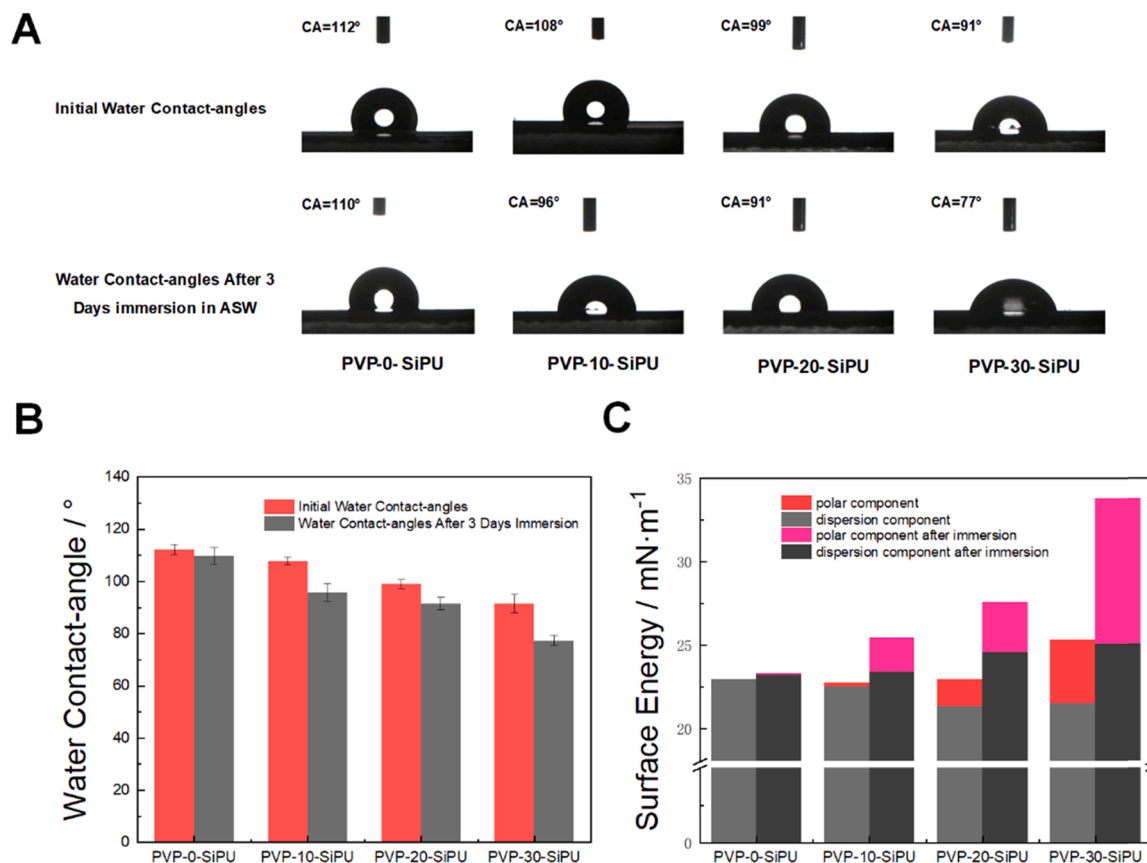


Fig. 3. (A) Photos of the water contact angles (WCAs) (B) Values of the WCAs (C) Surface energy of the coatings before and after immersion in artificial seawater for 3 days.

impact of currents in the real sea environment. As shown in Fig. 7, there was no damage such as swelling or peeling on the coating surface after being immersed in the real sea for 150 days (the small amount of attachment on surface was bryozoans, not damage of coating), which proved the application mechanical property of this low modulus antifouling coating in real sea.

3.5. Elastic Modulus Test

According to the quantitative relationship between fouling release performance and the elastic modulus (E), the lower the modulus of the low surface energy antifouling coating, the better the fouling release performance [3,4,23,50]. Inspired by the soft body of squid, reduction of surface modulus is feasible methods to enhance marine antifouling performance of marine equipment surface. PVP-OH is uniformly cross-linked in the coating to form random branched segments, which causes changes in the micromorphology of the coating and regulates the mechanical properties of the coating. PVP forms stable doping in the PDMS-based polyurethane by covalent bonds. Compared with the unmodified PDMS-based polyurethane coating PVP-0-SiPU, the elastic modulus of PVP-10-SiPU is reduced from 1.61 ± 0.12 MPa to 1.06 ± 0.17 MPa, getting a decrease of 34.16%, while the elastic modulus of PVP-20-SiPU and PVP-30-SiPU are 0.83 ± 0.08 MPa and 0.65 ± 0.11 MPa, respectively (Fig. 4). The results indicated that the elastic modulus of the hydrophilic modified coating doped with PVP-OH decreases, even lower than PDMS (~ 1 MPa), which is beneficial to reduce the settlement of fouling organisms on the coating surface. Part of hydrophilic PVP segment and the PTMG segment in polymer due to cause phase separation and form hydrophilic microdomains in the coating. These micro-regions (Fig. 2) destroyed the continuity of the PDMS coating to a certain extent. Since the mechanical strength of PVP and PTMG is lower than that of PDMS, these low modulus micro-domains have a significant effect on reducing the overall modulus of the polymer. However, the main body of the coating is still PDMS, and the low modulus area is only the dispersed micro areas, so the overall mechanical properties and swelling resistance of the coating will not be seriously impaired. In fact, after 150 days of immersion in the real sea, the surface of the coating was intact and there was no mechanical damage, which proved its excellent practical application performance.

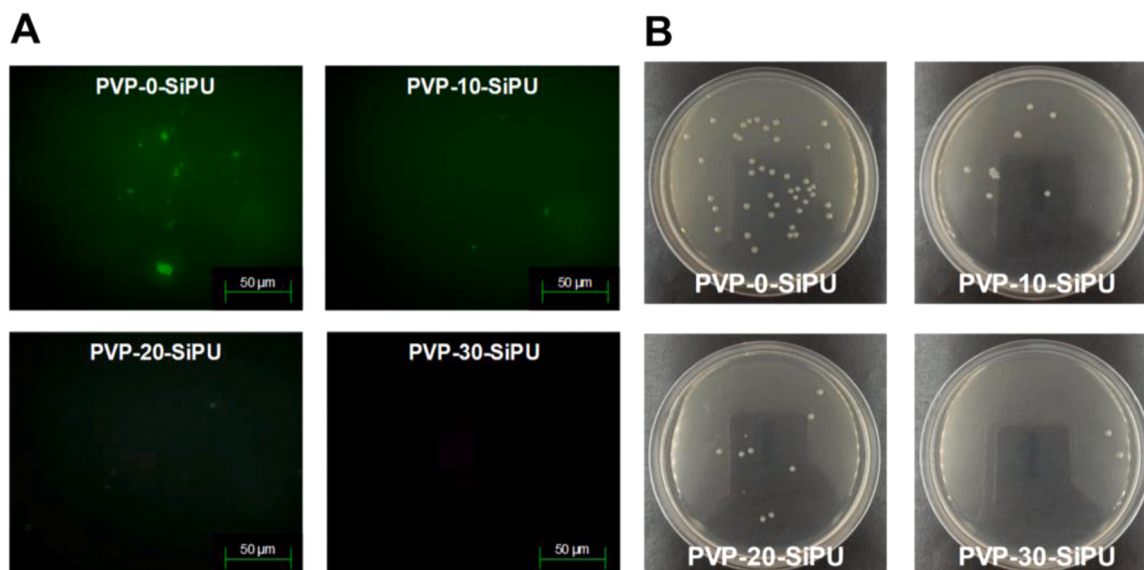


Fig. 5. (A) Anti-protein adsorption performance of the coating, fluorescein isothiocyanate labeled bovine serum albumin (FITC-BSA) as a model protein (B) Bacterial biofouling resistance test with *E.coli* after 5 h of contact with the sample PVP- x-SiPU.

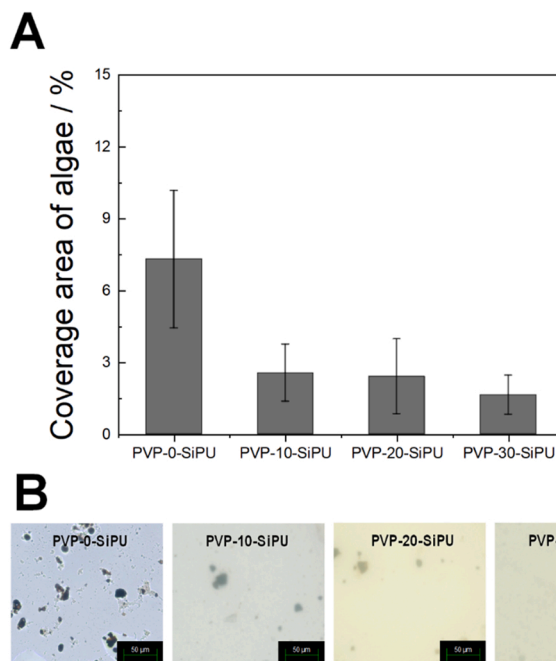


Fig. 6. Algae inoculation test after 7 days of culture (A) Percentage of algae coverage on different coatings and (B) Representative microscopic images of different coatings.

3.6. FITC-BSA Anti-adsorption, Bacteriostatic, Algae Plate Inoculation Experiments and Marine Field Antifouling Test

3.6.1. FITC-BSA Anti-adsorption test

The hydrophilic surface has an inhibitory effect on protein adsorption [51]. Herein, to anchor the hydrophilic PVP on the surface of the low surface energy coating, the triethoxy-containing silane coupling agent was reacted with the hydroxyl-containing PVP.

Although PDMS-based polyurethane coatings have lower surface energy, their hydrophobic properties are not conducive to the desorption of protein fouling by the coating [52]. Hydrophilic modification of low surface energy coatings introduced hydrophilic segments such as PVP and anchored them on the surface of the coating via chemical

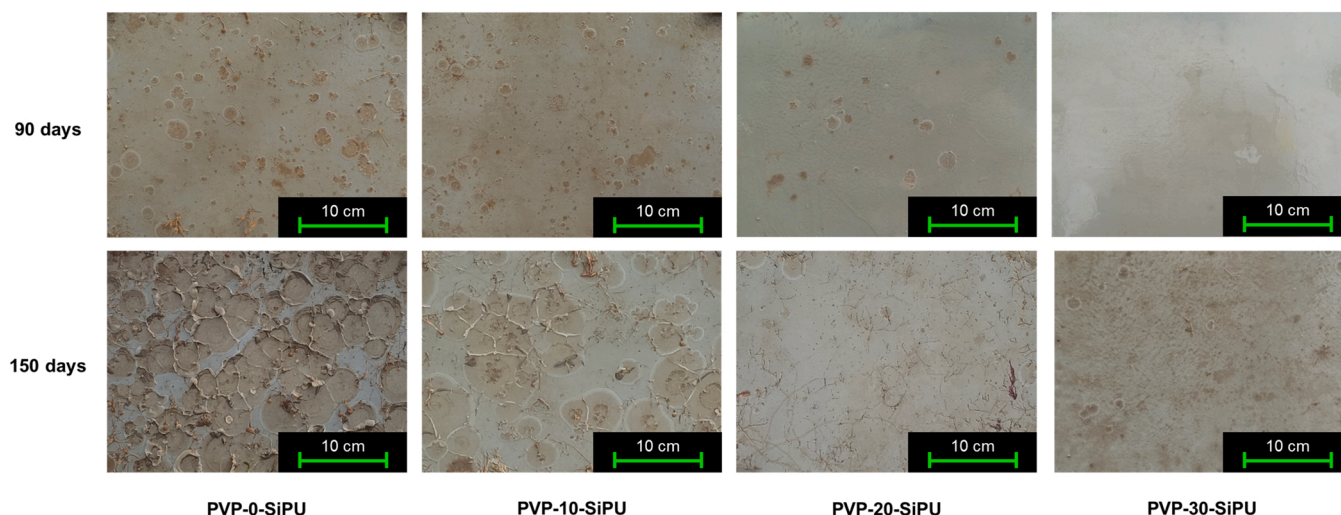


Fig. 7. Images of the tested panels coated with PVP-x-SiPU after immersion in seawater for 90 days and 150 days.

bonds. In an aqueous environment, water molecules and PVP form a hydration layer due to hydrogen bonding. The thermodynamic advantages of biological protein biomacromolecules in the adsorption process of the hydrophilic coating surface are greatly reduced so that it exhibits good anti-bioproduct properties. The hydration layer can prevent the adsorption of non-specific proteins, so it can endow the coating with long-term stable anti-protein adsorption and anti-microbial adhesion properties.

In the FITC-BSA anti-adsorption test, fluorescein isothiocyanate-labeled bovine serum albumin (FITC-BSA) was used as a model protein to investigate the anti-protein adsorption performance of the hydrophilic modified PDMS-based polyurethane antifouling coating. The test can simulate the antifouling performance of protein fouling in the early stage of marine fouling. Use an inverted fluorescence microscope to observe and photo the fluorescent response of the staining protein FITC-BSA. The different amounts of fluorescence in the photo represent different degrees of adsorption of the model protein on the coating.

As shown in Fig. 5(A), after the hydrophilic modification of the PDMS-based polyurethane, the hydrophilic modified coating had a significant anti-adsorption effect on BSA. When the PVP segment was not introduced, the PVP-0-SiPU coating adsorbed part of FITC-BSA, and it produced obvious fluorescence under a fluorescence microscope. The introduction of PVP enhanced the anti-protein adsorption performance of the hydrophilic modified coating. The surface of the PVP-10-SiPU coating only adsorbed trace amounts of BSA-FITC. The PVP-20-SiPU and PVP-30-SiPU coatings had almost no adsorption of BSA-FITC on the surface, indicating that the synthesized hydrophilic modified PDMS-based polyurethane coatings have good anti-protein adsorption properties.

3.6.2. Bacterial biofouling resistance test

Like squid and most fish producing hydrophilic mucus to protect the epidermis from bacteria and parasites, the hydrophilic modified coating also has bacterial biofouling resistance property. Due to the hydration layer on the coating surface, the hydrophilic modification of the surface further reduced the bacteria adsorption to the coating surface in an aqueous environment.

As shown in Fig. 5(B), the antibacterial properties of the hydrophilic modified PDMS-based polyurethane coating were studied by using the plate counting method and the addition of PVP-OH as the gradient. The antibacterial performance of the coating can be indirectly reflected by the number of colonies on the LB agar plate. Table S3 summarizes the average antibacterial efficiency of different coatings, the coating without hydrophilic modification was used as the control group. The

higher the amount of PVP added, the stronger the hydrophilicity of the coating, and the adhesion of bacteria was greatly reduced. The hydrophilic modified coating PVP-30-SiPU had the strongest degree of hydrophilicity, the best anti-bacterial adhesion effect, and the highest average antibacterial efficiency as high as 95.24%. The antibacterial performance test results showed that the antibacterial performance of the coating was improved after the introduction of PVP-OH. Based on the antibacterial performance test, the prepared low surface energy coating with tailored hydrophilic modification can inhibit the adhesion of bacteria on the surface during the initial stage of biofilm formation and has the potential for antibacterial applications in the field of medical devices.

3.6.3. Algae plate inoculation test

In the algae plate inoculation test, diatom was used to evaluate the antifouling performance of the coating against microscopic marine biofouling. Similar to the anti-bacterial adhesion mechanism, the hydration layer reduces the adhesion and growth of algae on the coating surface. As shown in Fig. 6, after the coating was put in the algae-containing artificial seawater for 7 days, the growth and colonization of diatoms on the coating surface and the formation of micro-fouling were observed under the microscope to verify the anti-algae and anti-fouling performance of the coating. The results showed that the introduction of the hydrophilic segment PVP-OH can reduce the growth and colonization degree of algae on the coating surface. Compared with the non-hydrophilic modified PVP-0-SiPU, the diatom coverage area of the coated PVP-10-SiPU surface was reduced by more than half. The degree of diatom fouling on the surface of PVP-10-SiPU, PVP-20-SiPU, and PVP-30-SiPU coatings was all maintained at a low level, and as the degree of hydrophilicity increased, the anti-algae performance was further improved.

3.6.4. Marine field antifouling test

To verify the real-sea antifouling performance of the hydrophilic modified PDMS-based polyurethane (PVP-x-SiPU) coating, the coated test panels were placed in the Zhoushan sea area of the East China Sea for 150 days. As the aforementioned analysis, the hydrophilic modification of the coating reduced the adsorption of primary fouling on the surface of the coating and realized the effective fouling release under the action of natural water currents in the marine environment. The elastic modulus of the modified coating was further reduced, and not easy to form marine fouling on the surface, besides, the excellent low modulus wasn't at the expense of durability during the marine field test. As shown in Fig. 7, the real sea hanging panels showed that after 150 days

of hanging in the sea, the degree of fouling on the surface became significant, and the surface of the unmodified coating PVP-0-SiPU has been fouled by marine organisms. The surface of the hydrophilic modified coating is relatively lightly fouled by marine organisms, and there is almost no obvious marine organism adhesion on the surface of PVP-30-SiPU. After the control sample (PVP-0-SiPU) was immersed in the sea for 150 days, a large number of bryozoans and a small amount of barnacles were attached to the surface. The photo of PVP-0-SiPU shows clear spots of bryozoans, but with the increase of PVP, the hydrophilicity of the surface increases, the modulus decreases, and the antifouling performance increases. The surface of PVP-30-SiPU has no obvious spots of bryozoans, which proves that the strategy of hydrophilization modification and modulus reduction can significantly enhance the real sea antifouling performance of low surface energy antifouling coatings.

The excellent performance of real sea antifouling performance comes from the hydrophilic defensive surface and low coating modulus. The hydrophilic modification of the low SE coating introduces hydrophilic segments, such as PVP, and fixes them on the surface of the coating through chemical bonds. In an aqueous environment, water molecules and PVP form a hydration layer due to hydrogen bonds. The highly hydrated surface of the hydrophilized polymer can bind water molecules so strongly that fouling organisms cannot replace these interface water molecules for adhesion. Therefore, it takes a lot of energy for fouling organisms to break through the hydration layer. The thermodynamic advantages of biofouling in the adsorption process of the hydrophilic coating surface are greatly reduced, so that it exhibits good antifouling performance. The hydration layer can give the coating long-term stable anti-protein adsorption and anti-microbial adhesion properties. Because the coating prevents protein, bacteria and algae from adhering to the surface, it delays the formation of the nutrient layer and microbial film on the surface, and prevents large fouling organisms such as barnacles and bryozoans from multiplying on the surface. At the same time, the larvae of large fouling organisms such as barnacles, mussels, bryozoans, etc. tend to attach to hydrophobic surfaces (like rocks) to enhance their own stability against waves. Therefore, the hydrophilic defensive surface can “confuse” these larvae and prevent them from attaching to the surface. Even if there are a small amount of larvae attached, it is easy to be detached under the washing of ocean waves due to the low modulus and low SE of the coating surface. The introduction of PVP-OH modified the low surface energy marine antifouling coating, conjointly increasing the hydrophilicity and reducing the elastic modulus, which is beneficial to improve the marine antifouling performance and demonstrate excellent practical application performance.

4. Conclusion

In summary, inspired by the soft body and epidermal mucus of squid, the stable PVP hydrophilic segments were introduced to construct a hydrophilic defensive surface on the PDMS-based polyurethane, realizing practical low surface energy coatings with jointly improved hydrophilicity and reduced elastic modulus. The existence of PVP segments played a critical role as an enhanced strategy for marine antifouling, in which hydrophilicity promoted the anti-adsorption effect on biological proteins and inhomogeneity by phase separation reduced the elastic modulus of the coating.

The anti-protein test was carried out with FITC-BSA, compared with the non-hydrophilic modified low surface energy coating, the bionic hydrophilic modified coating had excellent anti-protein adhesion performance. The introduction of the hydrophilic segment PVP-OH achieved reduced elastic modulus (<1.1 MPa), which is beneficial to the improvement of the antifouling effect. The marine field antifouling test showed that the low surface energy coating with tailored hydrophilic modification had a good marine antifouling performance as well as good application stability for 150 days in the marine environment, giving an efficient impetus to developing an eco-friendly and sustainable solution for no-antifoulant marine antifouling applications.

CRediT authorship contribution statement

Zheming Tong: Experiments and characterization operations; data curation; manuscript writing and revision; data supplementation; reply to reviewer comments. **Hongyu Guo:** Conceptualization and design of ideas. **Zhigang Di:** Validation of experiment procedure. **Ye Sheng:** Validation of experiment data. **Lina Song:** Resources (Provision of study materials). **Jiankun Hu:** Resources (Provision of study instrumentation). **Xiang Gao:** Review of draft and manuscript. **Yang Hou:** Review of figures and tables. **Xiaoli Zhan:** Project administration. **Qinghua Zhang:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.colsurfb.2022.112392.

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